

Class 15: Social networks and social capital

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Sociology 204: Social Networks
Princeton University

Wednesday, March 19, 2025



New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).

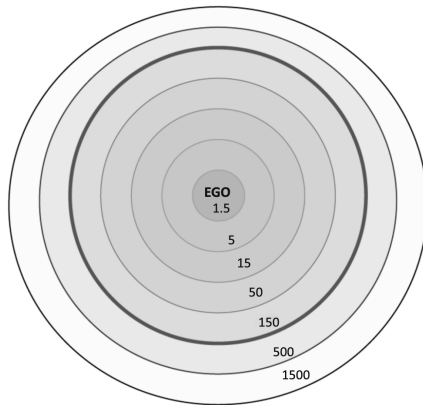
CORE DISCUSSION NETWORKS OF AMERICANS*

PETER V. MARSDEN

University of North Carolina at Chapel Hill

Aspects of interpersonal networks in which Americans discuss “important matters” are examined using data from the 1985 General Social Survey. These are the first survey network data representative of the American population. The networks are small, kin-centered, relatively dense, and homogeneous in comparison with the sample of respondents. Bivariate examination of subgroup differences by age, education, race/ethnicity, sex, and size of place indicates that network range is greatest among the young, the highly educated, and metropolitan residents. Sex differences consist primarily of differences in kin/nonkin composition of networks.

This week' assignment: core discussion networks of Princeton students. Start early.



Trends in Cognitive Sciences

Figure 2. The Circles of Friendship. An individual (ego) sits within a series of hierarchically inclusive circles, or layers, of friendship that have a very distinct scaling ratio (each layer is three times the size of the one inside it). The heavy line demarcates the 150 layer, the typical size of personal social networks. Beyond this are acquaintances (500 layer); the layer at 1500 appears to represent the limit on the number of faces we can put names to. Frequency of contact, rated emotional closeness, and willingness to act altruistically towards a given alter all decline across the successive circles [2,10].

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- ▶ weak ties also help hold communities together
- ▶ the relationship between tie strength and probability of finding a job is non-linear and depends on exactly how you measure tie strength

- ▶ Reeves and Fall (2023) Seven key takeaways from Chetty's new research on friendship and economic mobility Brookings Institution.
- ▶ Angrist and Sacerdote (2022) The social connections that shape economic prospects. Nature 608, 37-38.
- ▶ Chetty et al. (2022) Social capital I: measurement and associations with economic mobility. Nature 608, 108-121.
- ▶ Chetty et al. (2022) Social capital II: determinants of economic connectedness. Nature 608, 122-134.

Social capital I: measurement and associations with economic mobility

<https://doi.org/10.1038/s41586-022-04996-4>

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Raj Chetty^{1,7}✉, Matthew O. Jackson^{2,7}✉, Theresa Kuchler^{3,7}✉, Johannes Stroebel^{3,7}✉, Nathaniel Hendren¹, Robert B. Fluegge⁴, Sara Gong³, Federico Gonzalez⁴, Armelle Grondin⁴, Matthew Jacob⁴, Drew Johnston⁴, Martin Koenen⁴, Eduardo Laguna-Muggenburg⁵, Florian Muderekereza⁴, Tom Rutter⁴, Nicolaj Thor⁴, Wilbur Townsend⁴, Ruby Zhang⁴, Mike Bailey⁶, Pablo Barberá⁶, Monica Bhole⁶ & Nils Wernerfelt⁶

What is social capital?

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1. physical capital
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What is the most important kind of capital that you get from Princeton?

- ▶ Social capital is not just about individuals. You can benefit from the friendships that your classmates have.

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- ▶ If you want clearer definitions, I agree with you

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- ▶ “We define upward income mobility in each county as the average income percentile rank in adulthood of children who grew up in that county with parents at the 25th percentile of the national parental household income distribution”

What is economic mobility?

- ▶ “children’s chances of rising up the income distribution”
- ▶ “We define upward income mobility in each county as the average income percentile rank in adulthood of children who grew up in that county with parents at the 25th percentile of the national parental household income distribution”
- ▶ These estimates come from tax data and an earlier project (<https://www.opportunityatlas.org/>)

Measuring social capital

Measuring social capital

Building on previous work²⁷⁻²⁹, we organize measures of social capital into three categories: (1) cross-type connectedness, which is the extent to which different types of people (for example, high income versus low income) are friends with each other^{15,30-32}; (2) network cohesiveness, which is the degree to which friendship networks are clustered into cliques and whether friendships tend to be supported by mutual friends³³; and (3) civic engagement, which we measure using indices of trust or participation in civic organizations^{34,35}.

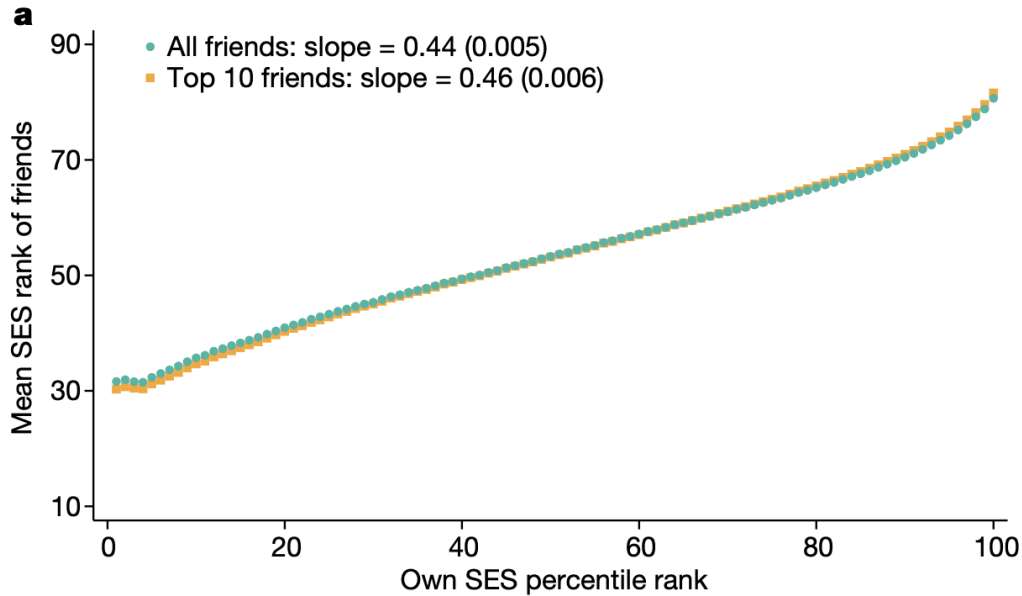
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First two are related to what Putnam calls “bridging” and “bonding” social capital

Economic connectedness

Many theoretical studies have shown how connections to more educated or affluent individuals can be valuable for transferring information, shaping aspirations and providing mentorship or job referrals^{15,30,31,38-44}. Consistent with these models, empirical studies have documented that social ties to well-resourced individuals can materially affect economic and labour market outcomes^{3-5,45}. Motivated by this literature, we begin by measuring connectedness across different types of people, focusing on economic connectedness: the extent to which people with low and high SES are friends with each other.



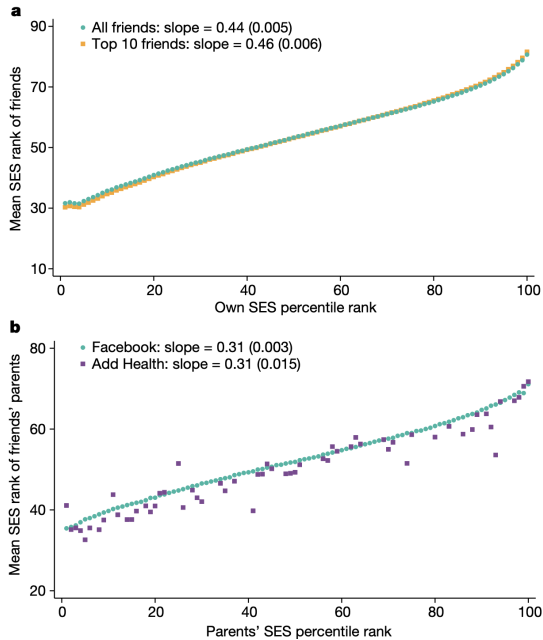
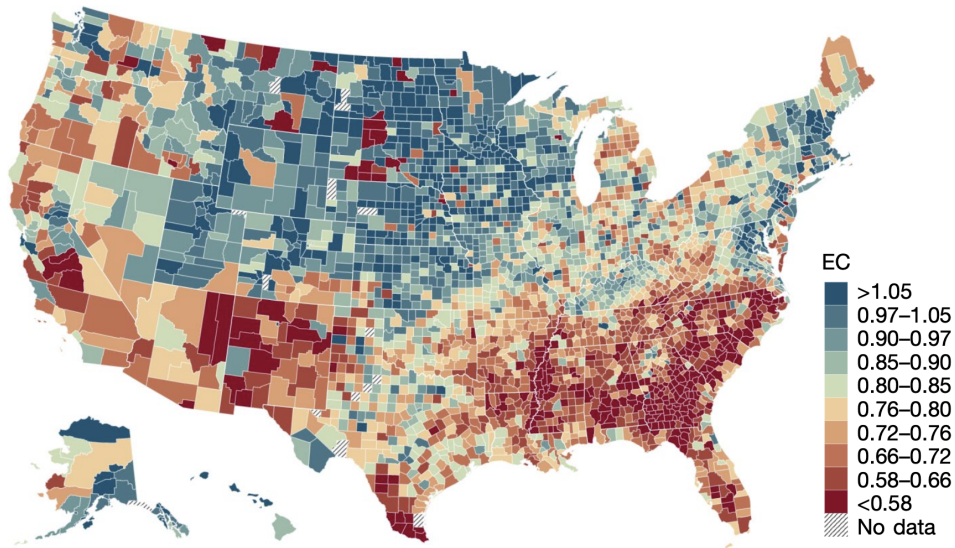


Fig. 1 | Relationship between an individual's SES and friends' SES.

edness'). A value of 0 for EC implies that a network has no connections between low-SES and high-SES people, whereas a value of 1 implies that low-SES people have an equal number of low-SES and high-SES friends.

a



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Fig2c: Clustering

c

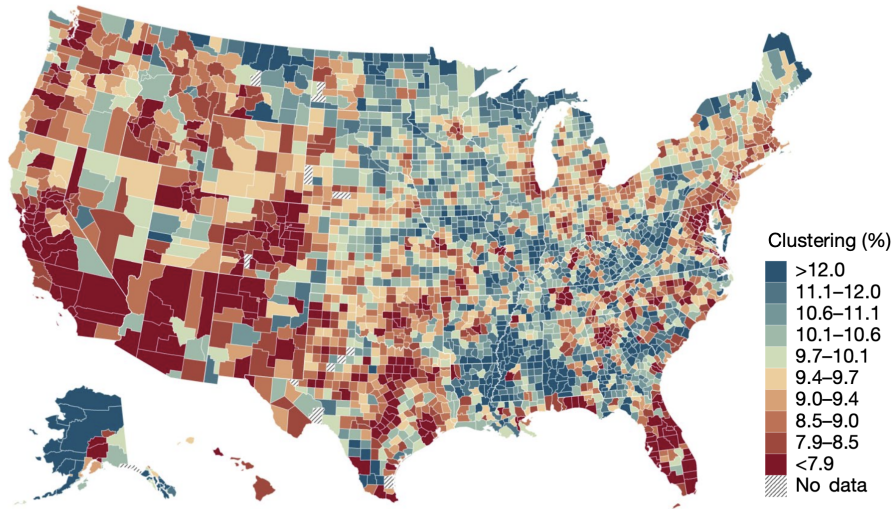
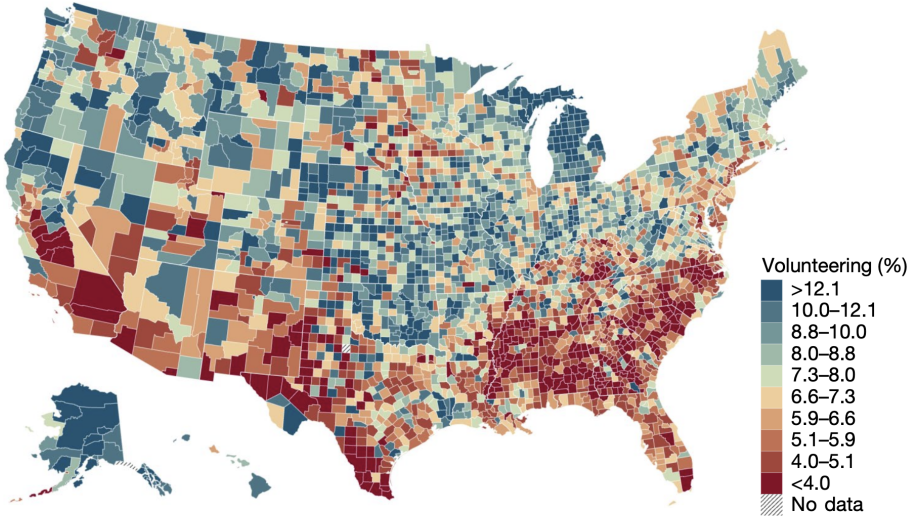
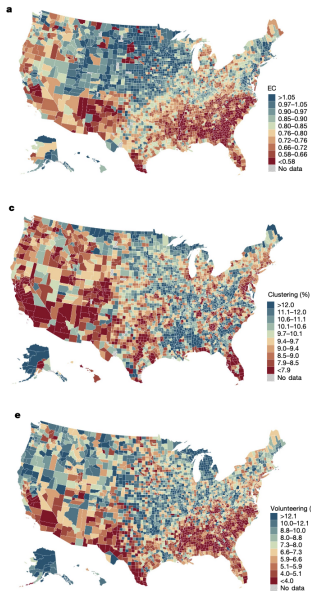


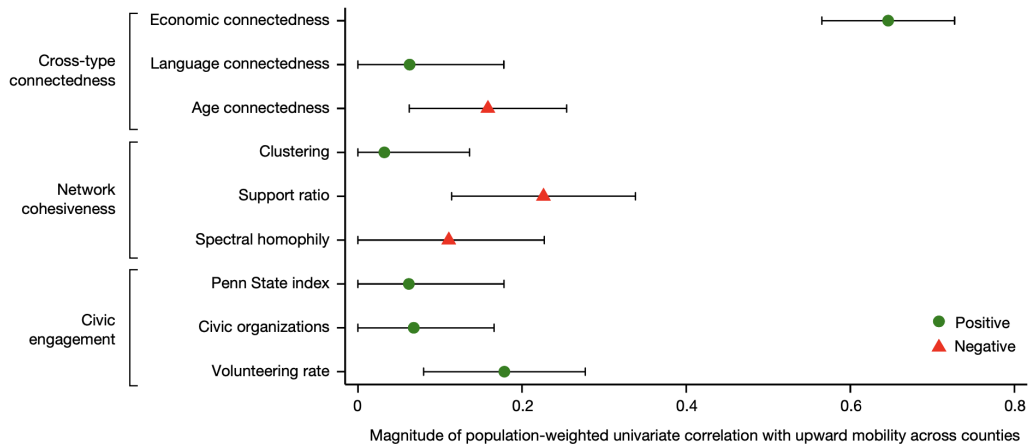
Fig2c: Volunteering rates

e

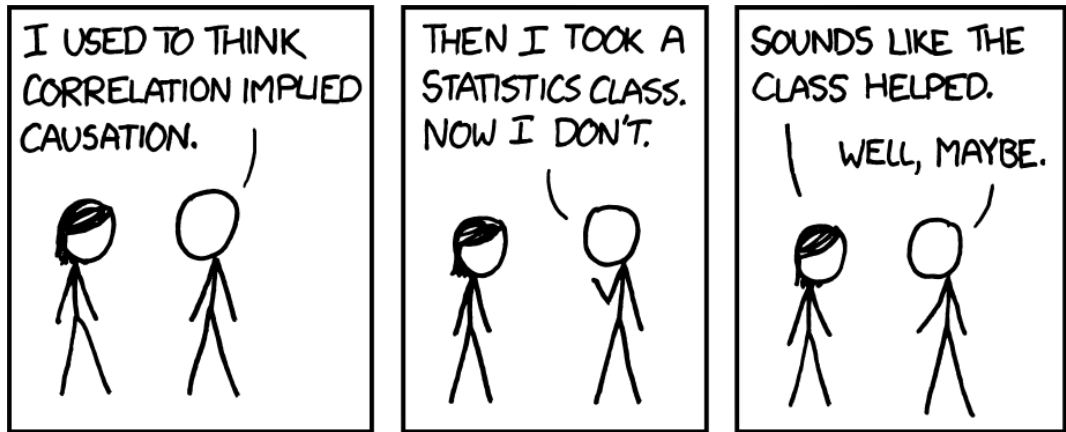


Which measure of social capital most associated with economic mobility?



a

The difference between what and why



The difference between what and why

Why EC is related to economic mobility

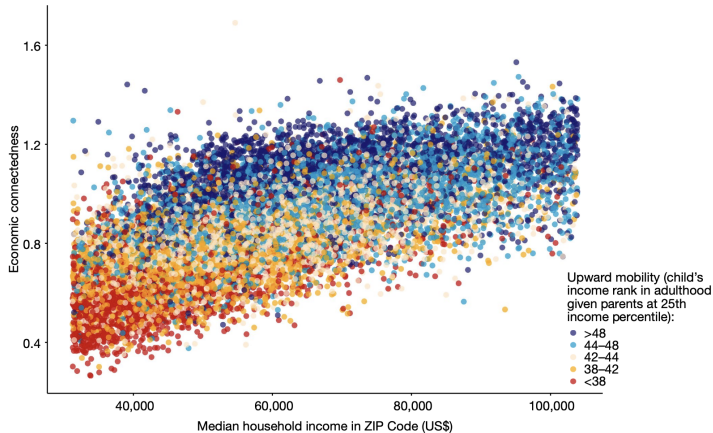
There are many theories for why economic connectedness could have a positive causal effect on upward income mobility. For example, economic mobility might be facilitated by connections to people who can shape aspirations or provide access to information and job opportunities^{30,32}. This interpretation is consistent with the argument that bridging capital—a concept that encompasses EC—is particularly valuable for ‘getting ahead’³⁶. However, there are also many alternative explanations for the correlation between EC and mobility that do not rely on a causal effect of connectedness on mobility. We evaluate three such possibilities in turn—reverse causality, selection effects, and omitted variables—with the broader aim of better understanding the channels through which connectedness and mobility are related.

The difference between what and why

Why EC is related to economic mobility

There are many theories for why economic connectedness could have a positive causal effect on upward income mobility. For example, economic mobility might be facilitated by connections to people who can shape aspirations or provide access to information and job opportunities^{30,32}. This interpretation is consistent with the argument that bridging capital—a concept that encompasses EC—is particularly valuable for ‘getting ahead’³⁶. However, there are also many alternative explanations for the correlation between EC and mobility that do not rely on a causal effect of connectedness on mobility. We evaluate three such possibilities in turn—reverse causality, selection effects, and omitted variables—with the broader aim of better understanding the channels through which connectedness and mobility are related.

They argue against each of these alternative explanations (pretty convincingly for me)



Roughly shows

- ▶ For a given level of median income in zip code, zip codes with more economic connectedness have higher upward mobility (vertical slices).
- ▶ For a given economic connectedness, zip codes with higher media income in zip code, have similar levels of upward mobility (horizontal slides).

Social capital I: measurement and associations with economic mobility

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Social capital—the strength of an individual’s social network and community—has been identified as a potential determinant of outcomes ranging from education to health¹⁻⁸. However, efforts to understand what types of social capital matter for these outcomes have been hindered by a lack of social network data. Here, in the first of a pair of papers⁹, we use data on 21 billion friendships from Facebook to study social capital. We measure and analyse three types of social capital by ZIP (postal) code in the United States: (1) connectedness between different types of people, such as those with low versus high socioeconomic status (SES); (2) social cohesion, such as the extent of cliques in friendship networks; and (3) civic engagement, such as rates of volunteering. These measures vary substantially across areas, but are not highly correlated with each other. We demonstrate the importance of distinguishing these forms of social capital by analysing their associations with economic mobility across areas. The share of high-SES friends among individuals with low SES—which we term economic connectedness—is among the strongest predictors of upward income mobility identified to date^{10,11}. Other social capital measures are not strongly associated with economic mobility. If children with low-SES parents were to grow up in counties with economic connectedness comparable to that of the average child with high-SES parents, their incomes in adulthood would increase by 20% on average. Differences in economic connectedness can explain well-known relationships between upward income mobility and racial segregation, poverty rates, and inequality¹²⁻¹⁴. To support further research and policy interventions, we publicly release privacy-protected statistics on social capital by ZIP code at <https://www.socialcapital.org>.

Where does economic connectedness come from?

Social capital II: determinants of economic connectedness

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In this Article, we analyse the determinants of social interactions across class lines, examining why people with higher SES tend to have more high-SES friends. Building on previous research^{9,10,11}, we distinguish two channels that can generate differences in an individual's share of high-SES friends: (1) differences in high-SES exposure, the share of high-SES members in the groups to which people with different socioeconomic statuses belong (for example, their schools or religious organizations), and (2) differences in friending bias, the rate at which people befriend high-SES individuals conditional on the share of high-SES members in the group (that is, conditional on exposure).

Two possible mechanisms:

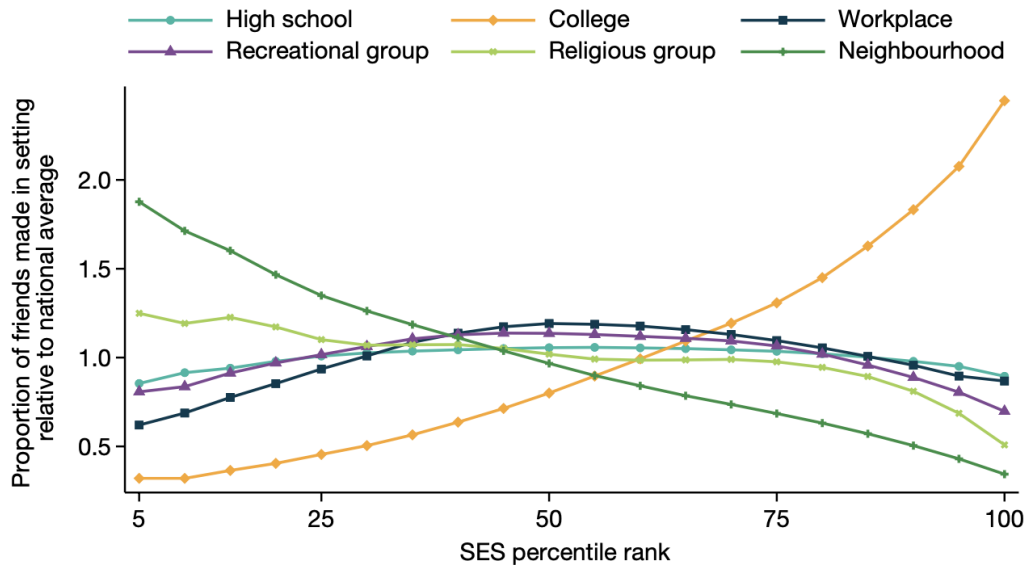
- ▶ exposure
- ▶ friending bias

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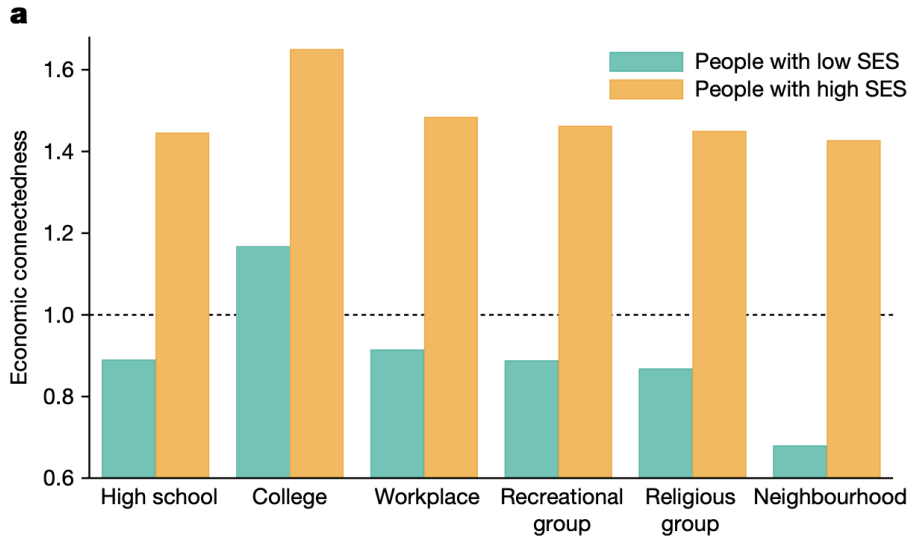
- ▶ exposure
- ▶ friending bias

This is not A or B. It is A and B.

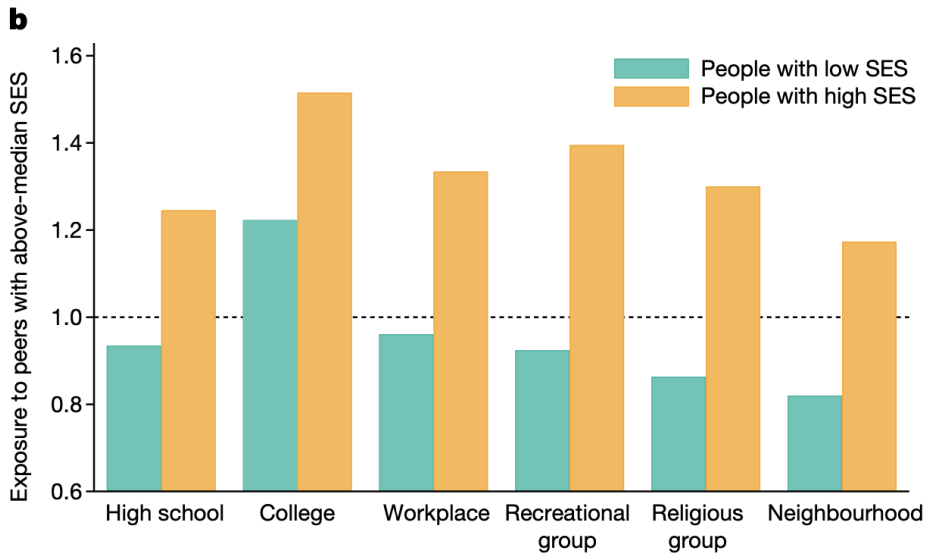
Where do friendships form?



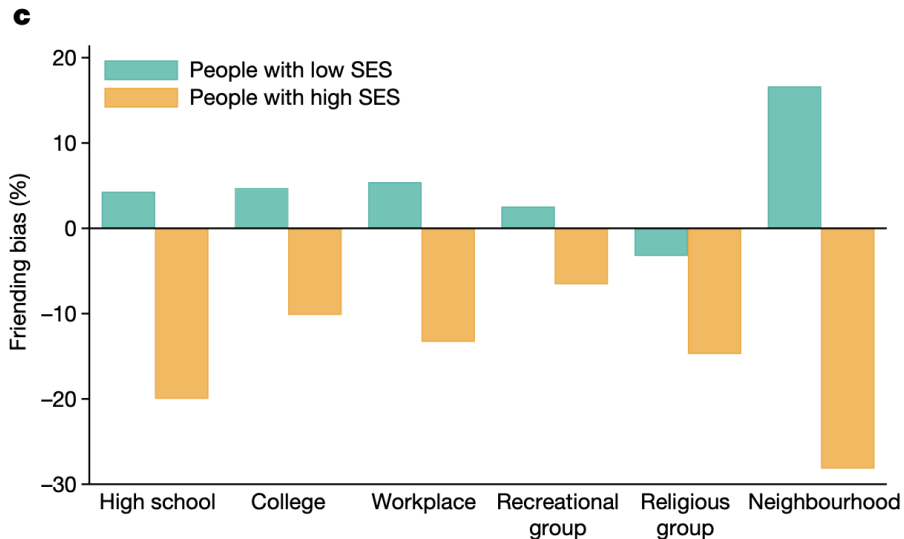
What settings lead to economic connectedness?



Exposure by setting

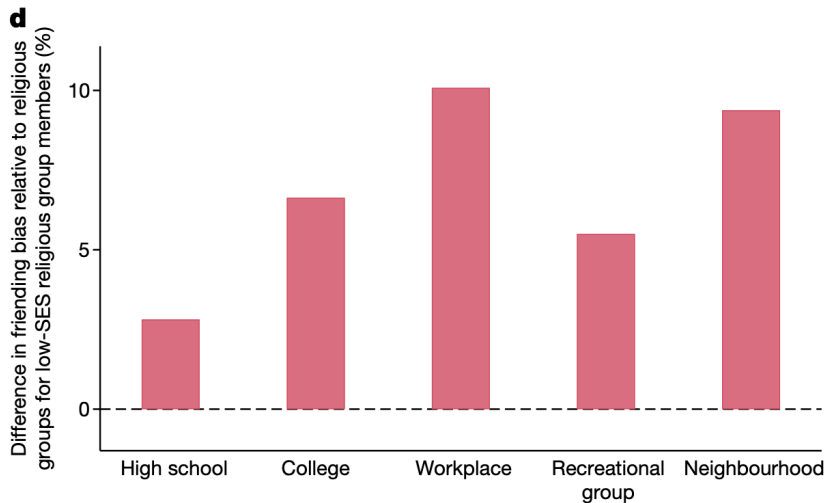


Friending bias by setting



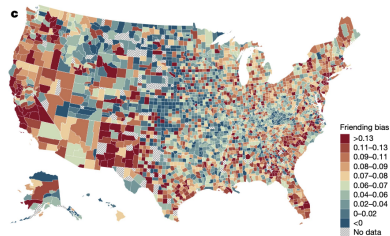
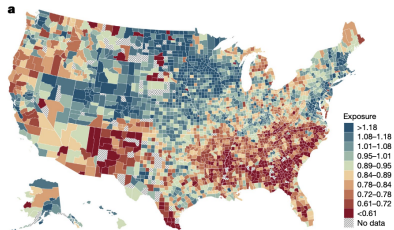
Note that religious groups are an outlier

Friending bias by setting for low SES people in religious groups

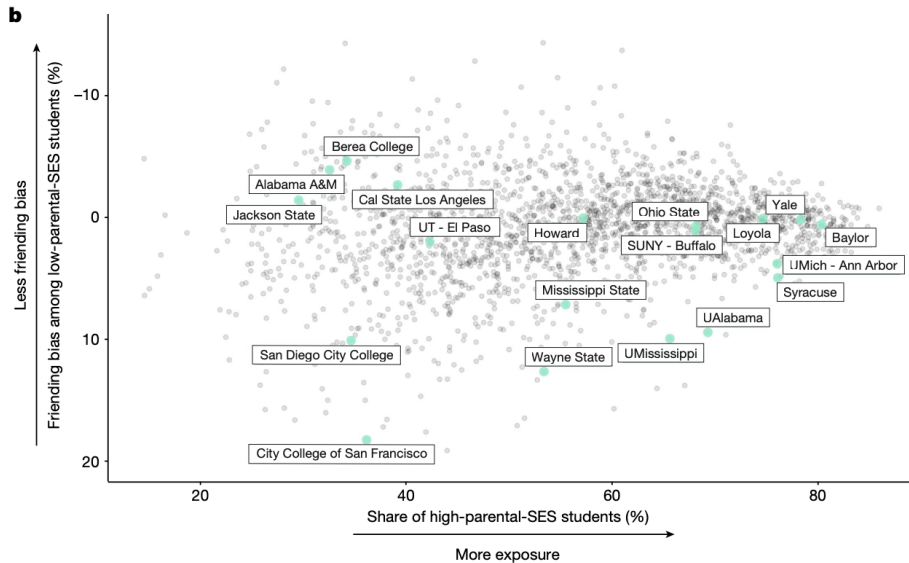


It is not the people, its the setting

Exposure and friending bias



Exposure and friending bias



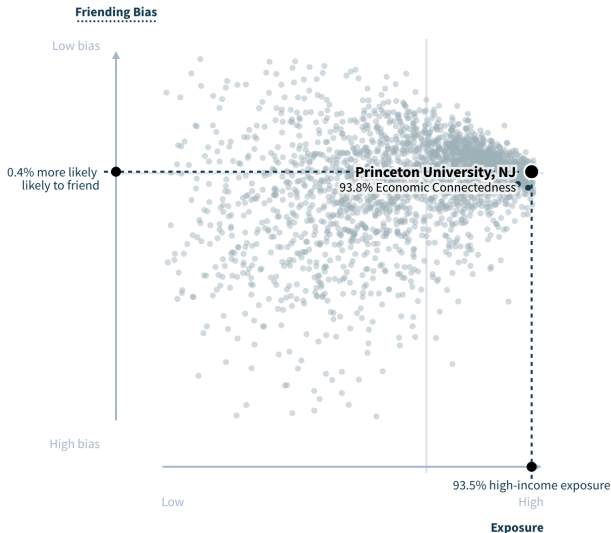
Exposure and friending bias, by college

Exposure and Friending Bias ⓘ

Scatter Plot

Maps

Strip Plots



PRINCETON UNIVERSITY, NJ

This college exhibits **high** exposure (99th percentile nationally), **average** friending bias (52nd percentile nationally) and **high** economic connectedness (99th percentile nationally) compared to the country as a whole.

☐ Show Outliers

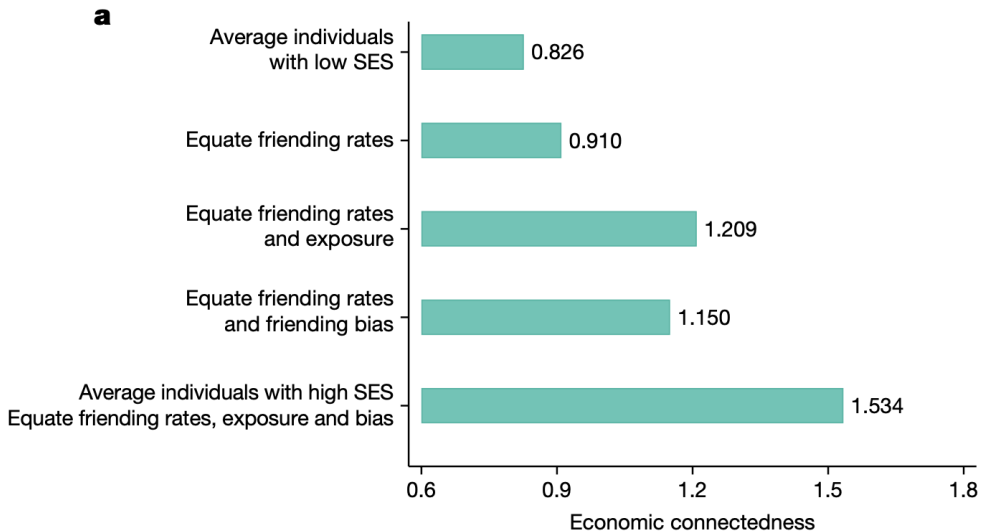
☐ Color by Economic Connectedness

What if?

Decomposing connectedness by SES

We quantify how much of the difference in the share of high-SES friends between people with low versus high SES is due to differences in friending shares across settings, differences in exposure, and differences in friending bias by conducting counterfactual exercises that sequentially remove variation in each of these three dimensions (Methods: 'Decomposing EC'). Conceptually, our goal is to determine how much of the difference in connectedness would remain if people with low SES made friends in different settings at the same rates as high-SES people (same friending shares); if they participated in groups with the same shares of high-SES members (same exposure); and if they made friends with high-SES peers at the same rates conditional on exposure as high-SES individuals do (same friending bias).

What if?



What if?

Through a series of analyses they argue that exposure bias and friending bias are shaped by the environment and therefore amenable to policy changes

What if?

- ▶ What are some policies that Princeton could do to increase exposure between low and high SES students?
- ▶ What are some policies that Princeton could do to decrease friending bias between low and high SES students?

Summary:

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- ▶ this relationship appears to be causal
- ▶ economic connectedness is related to exposure and friending bias
- ▶ exposure and friending bias vary across settings and may be amenable to policy interventions

Next class: a different view of on where friendships come from and how how life trajectories develop